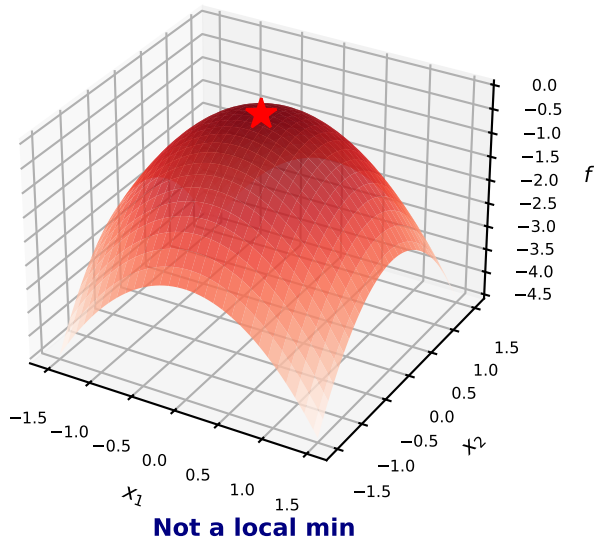
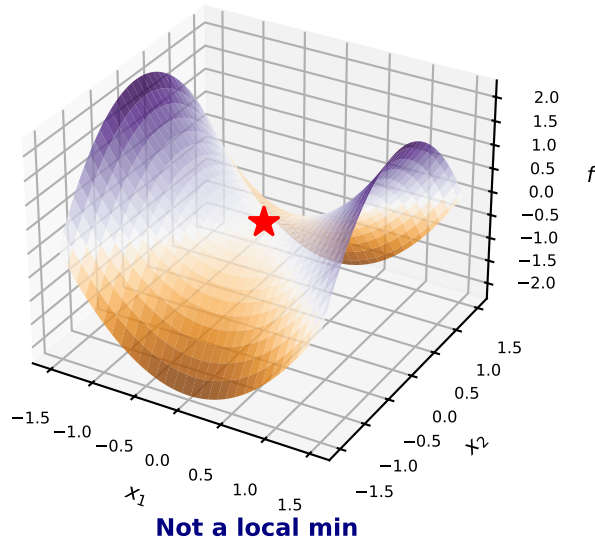


Three Local Regions Where $\nabla f = 0$ (Figure 1.15)

A Hill
 $\nabla f = 0$, $\nabla^2 f$ neg. def.



A Saddle
 $\nabla f = 0$, $\nabla^2 f$ indef.



A Bowl
 $\nabla f = 0$, $\nabla^2 f$ pos. def.

